

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 29)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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10	
11	
TOTAL	

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0 1 . 1

[6 marks]

Table 1

Method	Target species	One environmental disadvantage	
Demersal trawling			
Purse seining			
Pelagic long-lining			

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[2 marks]

0 1 . 3

Explain what is meant by **ghost fishing** and describe **two** ways it can be reduced.

[2 marks]

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0 2 . 1

Explain how overfishing of the giant triton mollusc damaged coral reefs in Australia.

[2 marks]

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0 2 . 2

Describe **four** management practices used in marine protected areas to protect coral reefs.

[4 marks]

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0 2 . 3

Explain why deep-water coral reefs recover much more slowly from trawling damage than tropical reefs.

[2 marks]

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0 2 . 4

Give **two** threats to deep-water reefs and **one** conservation measure.**[2 marks]****10**

0 3

Table 1 describes three genetically modified (GM) crops.

Table 1

GM crop	Gene source	Characteristic
Bt maize / cotton	Bacillus thuringiensis bacterium	Produces a toxin that kills insect pests
Roundup Ready soya	Agrobacterium	Resistant to the herbicide glyphosate
Golden rice	Daffodil and Erwinia bacterium	Produces vitamin A in the grain

0 3 . 1

Explain the main advantage of GM over conventional selective breeding.

[2 marks]

0 3 . 2

For each crop in Table 1, explain one potential benefit.

[3 marks]**Bt crops****Roundup Ready soya****Golden rice**

03 . **3** Explain **four** concerns about GM crops.

[4 marks]

03 . **4** Give **one** reason why GM crops have been slow to enter cultivation in Europe. **[1 mark]**

10

04 . **1** Explain the difference between hard and soft release and state the types of species for which each is suitable.

[3 marks]

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2

 Describe **five** features of a suitable release site.

[5 marks]

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0 4 . 3 Explain the importance of post-release monitoring.

[2 marks]

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0 5 . 1 Describe how soil texture is determined using a stack of sieves.

[3 marks]

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0 5 . 2 Explain why a soil auger gives a more representative sample than a trowel.

[2 marks]

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0 5 . 3

Explain why the organic matter content will be overestimated if a soil sample is heated above 550 °C.

[2 marks]

0 5 . 4

Describe how a turbidimeter measures water turbidity and explain how quantitative results are obtained.

[3 marks]

10

0 6 . 1

Explain why food surpluses produced in MEDCs in the 1970s could not easily be sold.

[3 marks]

0 6 . 2

Describe **four** methods used to reduce over-production while maintaining farm incomes.

[4 marks]

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0 6 . 3

Describe **three** features that farmers may be paid for under an agri-environmental scheme and the conservation purpose of each.

[3 marks]

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0 7

Table 1 shows some designated protected areas in the UK.

07 . 1

Complete Table 1 to show the legislation under which each area is designated and its key feature.

[4 marks]

Table 1

Designation	Designated under	Key feature	
Site of Special Scientific Interest (SSSI)			ox
Special Protection Area (SPA)			
Ramsar site			
Marine Conservation Zone (MCZ)			

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07 . 2

Explain how designating a protected area protects habitats and species.

[3 marks]

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07 . 3

Explain why voluntary agri-environmental schemes are needed in addition to legislation.

[3 marks]

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08 . 1

Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

08 . 2

Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

08 . 3

Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

5

09 . 1

Complete Table 1 to compare natural systems with typical linear human systems.

[4 marks]

Table 1

Feature	Natural systems	Linear human systems	
Energy source			ox
Use of materials			
Nature of wastes			
Temperature of processes			

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09 . 2

Explain why natural systems can be sustained using a relatively small number of elements.

[2 marks]

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09 . 3

Explain why the human use of minerals makes them difficult to recover and re-use.

[2 marks]

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Explain why toxic metals in electronic products cause problems that the wastes of natural systems do not.

[2 marks]

10

1 0 . 1

Explain how an understanding of ecological succession and population dynamics can inform conservation decision-making.

[9 marks]

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles.

[25 marks]

OR

1 1 . 2

Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS